

CS536 Science of Programming
Fall 2025
Assignment 4 Sample Solution Sketches

1. (a)

$$\begin{aligned} wlp(S, q) &\equiv wlp(x := x + 1; y := y \% x, \text{sqrt}(y) > x) \\ &\equiv wlp(x := x + 1, \text{sqrt}(y \% x) > x) \\ &\equiv \text{sqrt}(y \% (x + 1)) > x \end{aligned}$$

(b)

$$wlp(S, q) \equiv D(\text{sqrt}(y \% (x + 1)) > x) \Leftrightarrow x + 1 \neq 0 \wedge y \% (x + 1) \geq 0$$

(c)

$$\begin{aligned} D(S) &\equiv D(x := x + 1; y := y \% x) \\ &\equiv D(x := x + 1) \wedge wp(x := x + 1, D(y := y \% x)) \\ &\equiv T \wedge wp(x := x + 1, x \neq 0) \\ &\Leftrightarrow x + 1 \neq 0 \end{aligned}$$

(d)

$$\begin{aligned} wp(S, q) &\equiv (x + 1 \neq 0) \wedge (\text{sqrt}(y \% (x + 1)) > x) \wedge (x + 1 \neq 0 \wedge y \% (x + 1) \geq 0) \\ &\Leftrightarrow x + 1 \neq 0 \wedge \text{sqrt}(y \% (x + 1)) > x \wedge y \% (x + 1) \geq 0 \end{aligned}$$

2. (a)

$$\begin{aligned} wlp(S, q) &\equiv wlp(\text{if } y \geq 0 \rightarrow x := y/x \square x \geq 0 \rightarrow x := x/y \text{ fi}, x > y \wedge y > z) \\ &\equiv (y \geq 0 \rightarrow wlp(x := y/x, x > y \wedge y > z)) \\ &\quad \wedge (x \geq 0 \rightarrow wlp(x := x/y, x > y \wedge y > z)) \\ &\equiv (y \geq 0 \rightarrow y/x > y \wedge y > z) \wedge (x \geq 0 \rightarrow x/y > y \wedge y > z) \end{aligned}$$

(b)

$$D(wlp(S, q)) \equiv D((y \geq 0 \rightarrow y/x > y \wedge y > z) \wedge (x \geq 0 \rightarrow x/y > y \wedge y > z)) \Leftrightarrow x \neq 0 \wedge y \neq 0$$

(c)

$$\begin{aligned} D(S) &\equiv D(y \geq 0 \vee x \geq 0) \wedge (x \geq 0 \vee y \geq 0) \wedge (y \geq 0 \rightarrow D(x := y/x)) \\ &\quad \wedge (y < 0 \rightarrow D(x := x/y)) \\ &\Leftrightarrow (x \geq 0 \vee y \geq 0) \wedge (y \geq 0 \rightarrow x \neq 0) \wedge (x \geq 0 \rightarrow y \neq 0) \end{aligned}$$

(d)

$$\begin{aligned} wp(S, q) &\equiv (x \geq 0 \vee y \geq 0) \wedge (y \geq 0 \rightarrow x \neq 0) \wedge (x \geq 0 \rightarrow y \neq 0) \wedge (x \neq 0 \wedge y \neq 0) \\ &\quad \wedge (y \geq 0 \rightarrow y/x > y \wedge y > z) \wedge (x \geq 0 \rightarrow x/y > y \wedge y > z) \end{aligned}$$

3. Denote $IF \equiv \mathbf{if} \ x \geq 0 \rightarrow x := y + 1; z := x \ \square \ x \leq 0 \rightarrow y := x - 1; z := y \ \mathbf{fi}$.

$$\begin{aligned}
& aged(x = y, IF) = \{x, y, z\} \cap \{x, y\} = \{x, y\}. \\
& sp(x = y, IF) \\
& \equiv sp(x = y \wedge x = x_0 \wedge y = y_0 \wedge x \geq 0, \ x := y + 1; z := x) \\
& \quad \vee sp(x = y \wedge x = x_0 \wedge y = y_0 \wedge x \leq 0, \ y := x - 1; z := y) \\
& \equiv sp(x_0 = y \wedge x_0 = x_0 \wedge y = y_0 \wedge x_0 \geq 0 \wedge x = y + 1, \ z := x) \\
& \quad \vee sp(x = y_0 \wedge x = x_0 \wedge y_0 = y_0 \wedge x \leq 0 \wedge y = x - 1, \ z := y) \\
& \equiv (x_0 = y \wedge x_0 = x_0 \wedge y = y_0 \wedge x_0 \geq 0 \wedge x = y + 1 \wedge z = x) \\
& \quad \vee (x = y_0 \wedge x = x_0 \wedge y_0 = y_0 \wedge x \leq 0 \wedge y = x - 1 \wedge z = y) \\
& \Leftrightarrow (x_0 = y \wedge y = y_0 \wedge x_0 \geq 0 \wedge x = y + 1 \wedge z = x) \\
& \quad \vee (x = y_0 \wedge x = x_0 \wedge x \leq 0 \wedge y = x - 1 \wedge z = y)
\end{aligned}$$

4.

$$\begin{aligned}
& sp(y = x + 1, \ y := y + 1; \mathbf{if} \ x < 0 \mathbf{then} \ y := -y \ \mathbf{fi}) \\
& \equiv sp(y_0 = x + 1 \wedge y = y_0 + 1, \ \mathbf{if} \ x < 0 \mathbf{then} \ y := -y \ \mathbf{else} \ \mathbf{skip} \ \mathbf{fi}) \\
& \quad \# aged(y_0 = x + 1 \wedge y = y_0 + 1, \ \mathbf{if} \ x < 0 \mathbf{then} \ y := -y \ \mathbf{else} \ \mathbf{skip} \ \mathbf{fi}) = \{y\} \\
& \equiv sp(y_0 = x + 1 \wedge y_1 = y_0 + 1 \wedge y = y_1 \wedge x < 0, \ y := -y) \\
& \quad \vee sp(y_0 = x + 1 \wedge y_1 = y_0 + 1 \wedge y = y_1 \wedge x \geq 0, \ \mathbf{skip}) \\
& \equiv (y_0 = x + 1 \wedge y_1 = y_0 + 1 \wedge y_1 = y_1 \wedge x < 0 \wedge y = -y_1) \\
& \quad \vee (y_0 = x + 1 \wedge y_1 = y_0 + 1 \wedge y = y_1 \wedge x \geq 0)
\end{aligned}$$

5. (a) True. The validity of the triple follows immediately from the definition of $wp(S, q)$.
- (b) False. If $\sigma \models wlp(S, q)$, then $M(S, \sigma) - \perp \models q$, which means that it is possible that $\perp \in M(S, \sigma)$.
- (c) False. The definition of $wp(S, q)$ logically implies that if $\sigma \not\models wp(S, q)$, then $M(S, \sigma) \not\models q$.
- (d) True. The contrapositive of the statement is “If $M(S, \sigma) = \{\perp\}$, then $\sigma \models wlp(S, q)$ ”; this is true by the definition of the weakest liberal precondition.
- (e) True. For any state $\sigma \not\models wp(S, q)$, $M(S, \sigma)$ must consist of pseudo-states and/or states that satisfy $\neg q$. Thus, if $\sigma \not\models wp(S, q)$; in other words, $M(S, \sigma) - \perp \models \neg q$. Thus, we can get $\sigma \models \{\neg wp(S, q)\} S \{\neg q\}$.
6. (a) False. The definition of the strongest postcondition indicates that $\models \{p\} S \{sp(p, S)\}$ but not that $\models_{tot} \{p\} S \{sp(p, S)\}$.
- (b) False. The contrapositive of the statement is “If $\sigma \not\models p$, then $M(S, \sigma) - \perp \not\models sp(p, S)$ ”; it is not necessarily true since when $\sigma \not\models p$ we know nothing about $M(S, \sigma)$ and $sp(p, S)$.
- (c) False. By the definition of the strongest postcondition, if $\sigma \models p$ then $M(S, \sigma) - \perp \models sp(p, S)$.

- (d) True. Since σ is a well-formed state and $\sigma(p) \neq \perp \wedge \sigma \not\models p$, thus $\sigma \models \neg p$. By the definition of the strongest postcondition, we have $\sigma \models \{\neg p\}S\{sp(\neg p, S)\}$.
- (e) False. When $\sigma \not\models p$ we know nothing about $M(S, \sigma)$ and $sp(p, S)$.

7. (a)

$$\begin{aligned} p[y + z/x] &\equiv \left(w * x \neq 0 \wedge z \leq 2 \rightarrow f(w) > 0 \wedge \forall x. \exists y. 0 \leq y \leq x \right. \\ &\quad \left. \wedge f(w \div x) + y > f(z) \right) [y + z/x] \\ &\equiv w * (y + z) \neq 0 \wedge z \leq 2 \rightarrow f(w) > 0 \wedge \forall x. \exists y. 0 \leq y \leq x \\ &\quad \wedge f(w \div x) + y > f(z) \end{aligned}$$

(b)

$$\begin{aligned} p[x + z/w] &\equiv \left(w * x \neq 0 \wedge z \leq 2 \rightarrow f(w) > 0 \wedge \forall x. \exists y. 0 \leq y \leq x \right. \\ &\quad \left. \wedge f(w \div x) + y > f(z) \right) [x + z/w] \\ &\equiv (x + z) * x \neq 0 \wedge z \leq 2 \rightarrow f(x + z) > 0 \wedge \left(\forall x_0. \exists y. 0 \leq y \leq x_0 \right. \\ &\quad \left. \wedge f(w \div x_0) + y > f(z) \right) [x + z/w] \\ &\equiv (x + z) * x \neq 0 \wedge z \leq 2 \rightarrow f(x + z) > 0 \wedge \forall x_0. \exists y. 0 \leq y \leq x_0 \\ &\quad \wedge f((x + z) \div x_0) + y > f(z) \end{aligned}$$

(c)

$$\begin{aligned} p[x + y/z] &\equiv \left(w * x \neq 0 \wedge z \leq 2 \rightarrow f(w) > 0 \wedge \forall x. \exists y. 0 \leq y \leq x \right. \\ &\quad \left. \wedge f(w \div x) + y > f(z) \right) [x + y/z] \\ &\equiv w * x \neq 0 \wedge (x + y) \leq 2 \rightarrow f(w) > 0 \wedge \left(\forall x_0. \exists y_0. 0 \leq y_0 \leq x_0 \right. \\ &\quad \left. \wedge f(w \div x_0) + y_0 > f(z) \right) [x + y/z] \\ &\equiv w * x \neq 0 \wedge (x + y) \leq 2 \rightarrow f(w) > 0 \wedge \forall x_0. \exists y_0. 0 \leq y_0 \leq x_0 \\ &\quad \wedge f(w \div x_0) + y_0 > f(x + y) \end{aligned}$$

8. (a) One of the examples can be:

$$(x * y)[2 / x][4 / y] \equiv (x * y)[4 / y][2 / x]$$

They both are syntactically equivalent to $2 * 4$, so they are syntactically equivalent to each other. The key here is to choose an expression e containing no y and an expression e' containing no x .

(b) One of the counterexamples can be:

$$(x * y)[y / x][2 / y] \not\equiv (x * y)[2 / y][y / x]$$

Because $(x * y)[y / x][2 / y] \equiv (y * y)[2 / y] \equiv 2 * 2$ and $(x * y)[2 / y][y / x] \equiv (x * 2)[y / x] \equiv y * 2$, so they are not syntactically equivalent to each other. The key here is to choose an expression e containing y and/or an expression e' containing x .